

LinkML Modeler

A visual schema editor for authoring LinkML models on an ERD-style canvas — without hand-editing YAML

Adam Labadorf^{1,2,3}

¹ Department of Neurology, Chobanian & Avedisian School of Medicine, Boston University ² Bioinformatics Program, Boston University ³ National Center for PTSD Brain Bank, U.S. Department of Veterans Affairs

React 18 · TypeScript
ReactFlow · Vite

The Problem

LinkML is a modeling language for describing data schemas as structured YAML. As schemas grow, hand-editing that YAML becomes error-prone, hard to review, difficult to reason about, and less accessible to non-technical data modelers.

LinkML Modeler turns the schema into a direct-manipulation diagram: classes and enums become nodes on an entity-relationship canvas, edited visually and exported back to clean LinkML YAML. UI has many features designed to aid in the creation and modeling of large schemas that span multiple LinkML YAML files.

Validation & Correctness

Structural meta-model checks run continuously, surfacing errors and warnings before export.

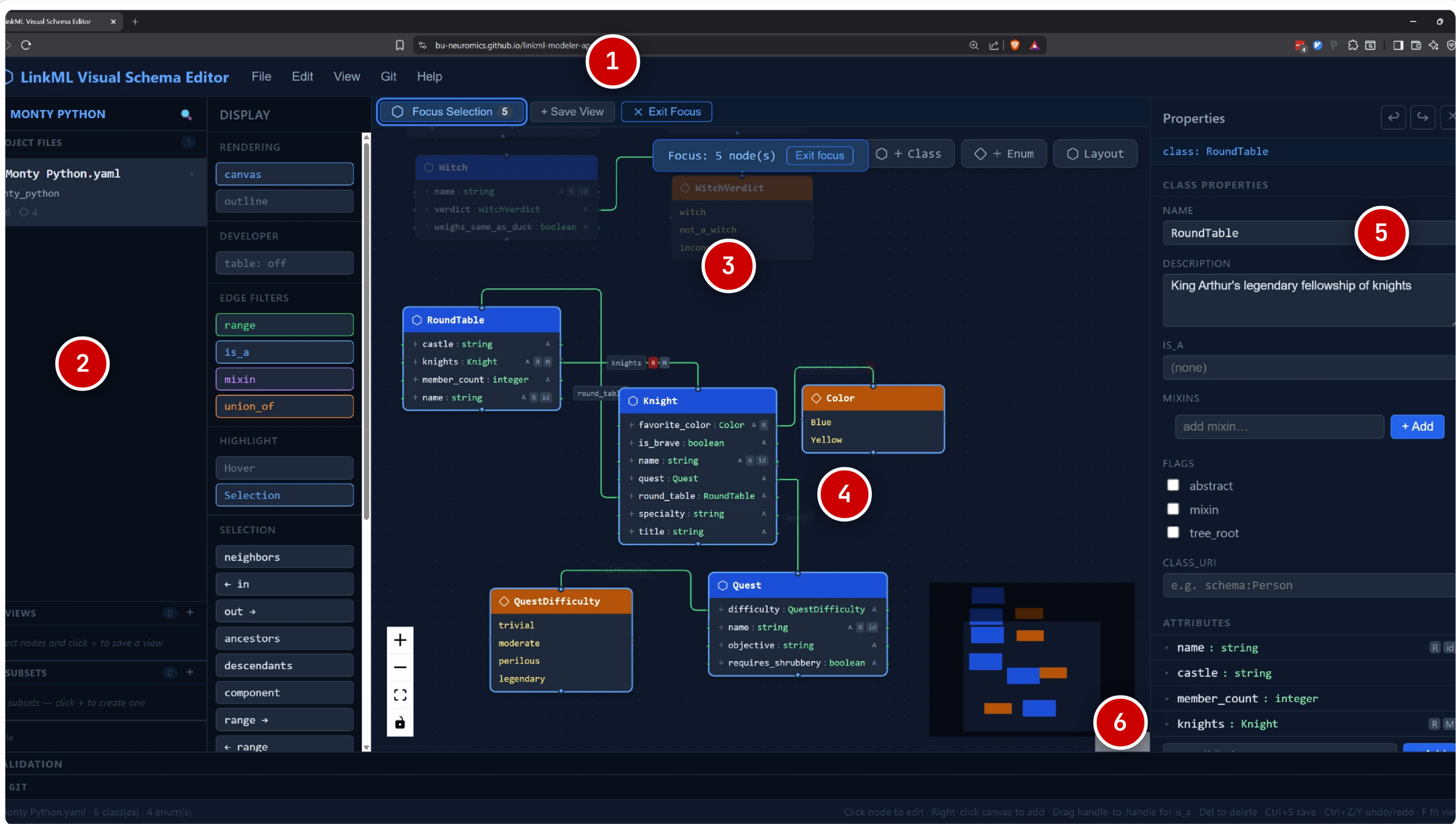
- Aa

Naming conventions
PascalCase classes & enums, snake_case slots
- ↻

Reference existence
is_a / mixin / range targets must resolve
- 🔄

Inheritance cycle detection
Catches circular is_a hierarchies
- 📄

Schema metadata
Required schema-level fields are checked



The visual ERD canvas

drag · connect · arrange · validate

Try it now

Runs entirely in the browser — no install required.



LIVE DEMO
bu-neuromics.github.io/
linkml-modeler-app/app



SOURCE
github.com/
BU-Neuromics

Anatomy of the editor

- 1

Toolbar
Focus mode, auto-layout, Git sync, and live YAML export.
- 2

Multi-schema selector
Multiple schema files listed for navigating large multi-component schemas
- 3

Class & enum nodes
Slots and ranges shown inline; color-coded by entity type.
- 4

Relationship edges
Drag between nodes to define is_a, mixins, and slot ranges.
- 5

Properties panel
Structured editor for slot definitions, ranges & constraints.
- 6

Minimap & zoom
Navigate large schemas; system-aware light & dark themes.